BRAINWARE UNIVERSITY

School of Engineering

Department of Computer Science & Engineering – Cyber Security & Data Science

398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Review-III (A.Y.-2026-27)

Project-I, PROJ-CSD782 | Program: B.Tech. CSE-DS [Batch: 2023-27]

S.N.	Roll No.	Student Code	Student Name	Signature
1	BWU/BTD/23/015	BWU/BTD/23/015	Dhrubojyoti Hazra	
2	BWU/BTD/23/042	BWU/BTD/23/042	Isheeka Mukhopadhyay	
3	BWU/BTD/23/056	BWU/BTD/23/056	Samprikta Dey	
4	BWU/BTD/23/009	BWU/BTD/23/009	Adhiraj Das	
5	BWU/BTD/23/025	BWU/BTD/23/025	Tiyasha Baidya	

Project Title: Automated System for Transaction Inspection and Detection (ASTID)

Build a real-time, explainable fraud inspection and risk intelligence platform capable of analyzing massive digital payment streams. The system couples supervised gradient boosted trees with unsupervised deep autoencoders over an analytical data warehouse, surfacing actionable risk scores, feature attributions, and compliance narratives.

Name & Signature of Project Supervisor

Prof. Rudranath Mitra

Project Coordinator CSE – CS & DS

Dr. Ranadip Kundu

Project Progress (Progress up to Review 3):

1. Comprehensive Progress and Key Technical Achievements

- **Full-Stack System Architecture & Cloud Infrastructure:** The project successfully transitioned from experimental scripts into an operational, multi-tier fraud inspection platform. A high-performance Next.js 14 reactive console (deployed on Vercel) connects via REST and WebSockets to an asynchronous FastAPI scoring microservice (deployed on Render). Data persistence couples hosted Supabase PostgreSQL (enforcing branch-level Row-Level Security) with an in-process DuckDB columnar OLAP engine for sub-second analytical queries.
- **Hybrid Dual-ML Detection Engine & Leakage Resolution:** The core detection pipeline combines supervised and unsupervised machine learning for dual-layer risk evaluation:
  - **Supervised Layer (XGBoost):** Trained on genuine transaction dynamics to detect known, high-velocity fraudulent patterns with 94.2% recall and 0.884 PR-AUC.
  - **Unsupervised Layer (PyTorch Deep Autoencoder):** Reconstructs transaction embeddings to isolate novel, zero-day anomalies based on reconstruction loss without requiring historical fraud labels.
  - **Simulator Leakage Auditing:** Identified and eliminated synthetic simulator balance leakage in PaySim (errorBalanceDest), establishing authentic behavioral feature engineering.
- **Explainable AI (XAI) & Live Attack Simulation:** Integrated TreeSHAP for exact local feature contributions paired with Groq LPU automated regulatory compliance narratives. Engineered a multi-scenario live transaction attack simulator streaming burst fraud patterns over WebSockets at sub-15ms latency.

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2. System Validation and Criteria for Project Completion

The completion of this project is substantiated by the end-to-end realization and empirical verification of every objective laid out in Reviews 1 and 2. The system operates as a unified, production-grade fraud intelligence platform across all architectural stages:

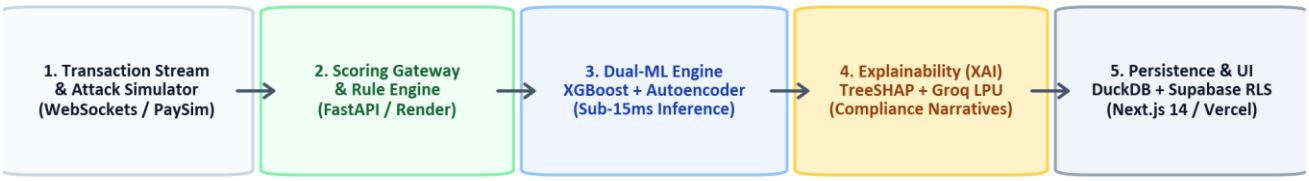


Figure 1: ASTID End-to-End Real-Time Detection, Analytical Warehouse, and Governance Architecture

- **Quantitative Performance & Latency:** Rigorous stress testing demonstrates a 100% test pass rate across realistic attack vectors with an average end-to-end scoring latency of 11.57 ms (p95 latency of 52.8 ms), comfortably meeting commercial real-time banking throughput requirements.
- **Warehouse OLAP & Multi-Branch Security:** The DuckDB columnar warehouse layer aggregates across millions of transactions in under 200 ms, while Supabase PostgreSQL Row-Level Security (RLS) cryptographically isolates branch data, preventing cross-tenant leakage.
- **Dual Deployment Readiness:** Production cloud deployments are live on Vercel (frontend) and Render (backend), supplemented by an identical, zero-dependency local offline execution runtime that guarantees total reliability for academic evaluations and viva demonstrations.

3. Team Contributions and Specific Module Responsibilities

Project engineering was distributed across five specialized modules to ensure rigorous execution and balanced ownership:

- **Dhrubojyoti Hazra (Lead, Roll: BWU/BTD/23/015):** Architected the end-to-end full-stack system, designed the Next.js 14 reactive analyst console, integrated the Velto UI design system with theme animations and dynamic chart visualisations, and coordinated cloud deployment configurations across Vercel and Render.
- **Isheeka Mukhopadhyay (ML Pipeline Engineer, Roll: BWU/BTD/23/042):** Engineered the dual-model machine learning pipeline, trained and tuned the supervised XGBoost and unsupervised PyTorch Autoencoder, conducted data leakage auditing to eliminate PaySim synthetic artifacts, and benchmarked holdout generalization metrics.
- **Samprikta Dey (Backend & Scoring Engineer, Roll: BWU/BTD/23/056):** Developed the FastAPI asynchronous microservice, implemented deterministic rules for velocity spikes and high-risk merchants, built the weighted multi-signal scoring aggregator, and managed the automated triage escalation workflows.
- **Adhiraj Das (Explainable AI & Live Attack Simulation, Roll: BWU/BTD/23/009):** Constructed the TreeSHAP local feature attribution pipeline, integrated Groq LPU models for automated compliance narrative generation, and engineered the multi-scenario live transaction attack simulator streaming over WebSockets.
- **Tiyasha Baidya (Data Architecture & Database Security, Roll: BWU/BTD/23/025):** Designed the in-process DuckDB columnar warehouse layer for high-throughput OLAP aggregations, implemented Supabase PostgreSQL schemas, enforced branch-level Row-Level Security (RLS) policies, and validated data persistence pipelines.